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DATA VISUALIZATION

Report Lab 03

DATA VISUALIZATION ON POWER BI

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The topic of analyzing and visualizing data from the Steam platform required not only data processing and analysis skills but also a clear understanding of how to present information in a visual and effective way. Thanks to our supervisor's guidance, we were able to approach the project step by step, from defining the problem and building analytical questions to selecting suitable charts and designing an appropriate dashboard.

In addition, the detailed feedback we received during the process helped us recognize our limitations in both design thinking and information organization. As a result, we were able to improve the quality of our project in terms of both content and presentation.

Although we have tried our best to complete this project, we understand that there are still areas that need further improvement. We sincerely look forward to receiving more feedback from our supervisor so that we can continue to improve in future projects.

Thank you very much.

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1 Information of Group 04

Table 1.1: Group Information and Individual Contribution Percentages

STT	Full name	MSSV	Specific tasks	Contributions
1	Le Tuan Anh	23127151	Objectives 4, 5, 6	100%
2	Nguyen Hoang Gia Bao	23127157	Objectives 7, 8, 9, 10	100%
3	Tran Huu Loc	23127407	Objectives 1, 2, 3	100%
4	Nguyen Thanh Loi	23127408	Objectives 11, 12, 13 (Machine learning)	100%

2 Learning Power BI

2.1 Overview of Power BI

Microsoft Power BI is a powerful data visualization tool developed by Microsoft. It allows users to connect, process, and visualize data from different sources. The tool supports creating interactive charts, dashboards, and reports, which help users explore data and make better decisions based on the information.

2.2 Main components of Microsoft Power BI

2.2.1 Power BI Desktop

- **Functions:** It is a free desktop application that offers a comprehensive environment for data integration, data cleaning, and data modeling
- **Applications:** It is mainly used by data analysts to develop complex data models and design detailed reports before publication.

2.2.2 Power BI Service

- **Functions:** It is a cloud-based software service (SaaS). It acts as a platform to store, manage, and share data visualization products.

- **Applications:** It is used to publish reports from Microsoft Power BI Desktop, create dashboards, manage access permissions, share data securely within an organization, and view reports online.

2.2.3 Power BI Report

- **Functions:** It is a collection of multiple pages that include charts, graphs, and tables, all built from a single data model.

- **Ứng dụng:** It provides users with a deeper and more detailed view of the data, allowing them to interact with it, apply filters, and explore specific metrics.

2.3 Basic functions of Microsoft Power BI

- **Import data:** Users can connect to and import data into Microsoft Power BI using the "Get Data" feature. This allows them to collect data from many different sources, such as local files (Excel, CSV, PDF), databases (SQL, Oracle), or web services.

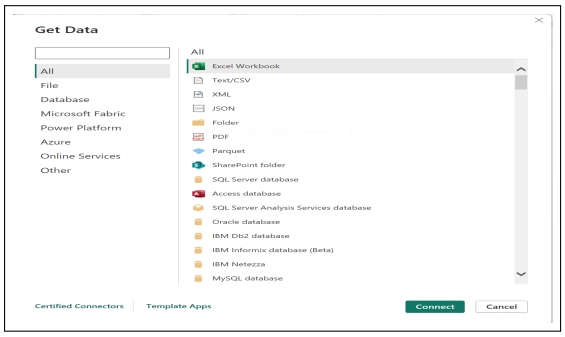
- **Data Transformation (Power Query):** Basic steps for cleaning and transforming data are performed in the Power Query Editor of Microsoft Power BI. Raw data is standardized through actions such as changing data types, removing missing or incorrect values, splitting columns, or merging tables. This process prepares the data for further analysis.

- **Data Modeling:** Users can create relationships between tables by defining keys in Microsoft Power BI. These relationships can be one-to-many or one-to-one. This helps build a logical data model, allowing data from different tables to filter and interact with each other smoothly.

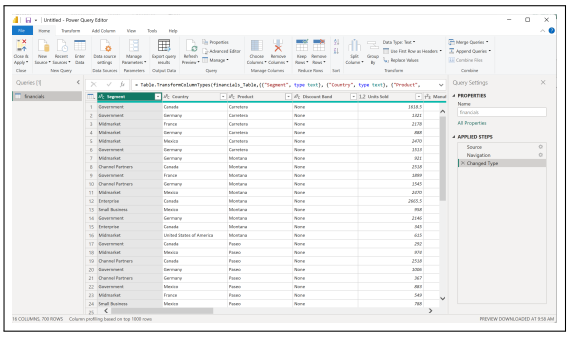
- **Visualization:** Users can create basic charts by dragging and dropping data fields into the Visualizations panel in Microsoft Power BI. The tool provides a wide range of chart types and allows users to customize formats and colors. It also supports filters and slicers, making it easier for users to interact with the data.

- **Dashboard:** Users can design and organize a dashboard layout directly in Microsoft Power BI Desktop by placing important charts, key performance indicators (KPIs), and filters on a single page. This design provides an overall view of the data and,

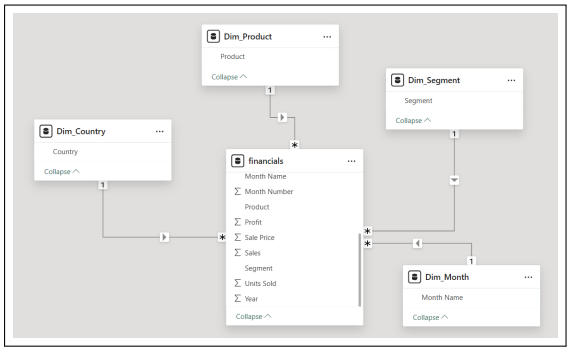
together with cross-filtering features, helps users quickly understand and interact with the information without switching between pages.



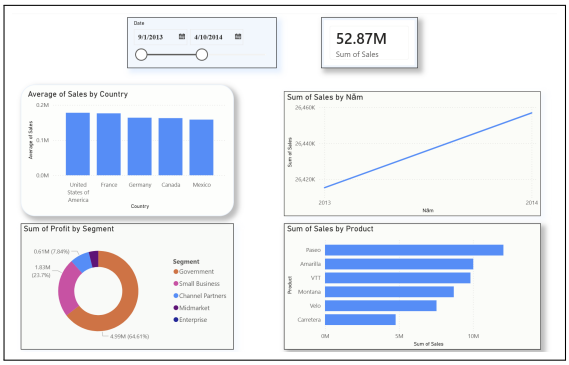
(a) Import data



(b) Power Query interface



(c) Relationship Diagram



(d) Dashboard

Figure 2.1: Functions of Microsoft Power BI

3 Introduction to the analysis problem

3.1 Overview of the problem

The game industry on the Steam platform is growing rapidly, with increasing competition. Thousands of new games are released every year, making it more difficult for developers, especially indie studios, to position their products, reach players, and develop effective strategies.

To study this market, our group used the dataset **Steam Games, Reviews, and Rankings** collected from Kaggle. This dataset includes more than 990,000 real user reviews for 242 games, along with detailed information about 290 top games based on sales, revenue,

and ratings across six main genres. The data is divided into three main tables, which are connected through the `game_name`:

- **games_description.csv:** It contains detailed information about each game, such as the name, developer, publisher, release date, genre, and system requirements.
- **steam_game_reviews.csv:** It contains user reviews, including review content, playtime, helpful votes, and recommendation status.
- **games_ranking.csv:** It classifies game rankings within each genre based on criteria such as sales, revenue, and user reviews.

3.2 Overall analysis objectives

"What makes a game successful on Steam? It could be the market, the player community, the developer's strategy, or the pricing of the product.?"

3.3 Specific analysis objectives

To address the overall problem, the group divided the analysis process into 13 specific objectives, covering four main aspects of the market:

- **Objective 1:** How does the number of games released on Steam change each year, and has the market become saturated yet?
- **Objective 2:** Which game genres account for the largest share, and how has the distribution of genres changed over time?
- **Objective 3:** Which developers contribute the largest number of games, and what is the quality of their games?
- **Objective 4:** Which publishers have the most games and the highest total number of reviews, and who is dominating Steam?
- **Objective 5:** How do indie games and publisher-backed games differ in terms of user engagement and recommendation review rates on Steam?
- **Objective 6:** Is the level of game globalization (measured by the proportion of non-English reviews) positively correlated with the total number of reviews and the positive rating rate on Steam?

- **Objective 7:** Is the number of "helpful votes" on reviews correlated with the positive review rate on Steam?
- **Objective 8:** Analyzing Different Trends Between Negative and Positive Reviews
- **Objective 9:** Analyzing Player Satisfaction by Game Genre
- **Objective 10:** Correlation Between Engagement Volume and Satisfaction Level
- **Objective 11:** Do top-ranking games by Revenue have higher average playtime and recommendation rates compared to other ranking types (Sales/Review)?
- **Objective 12:** Which month of the year receives the most reviews—is there a 'golden season' for releases?
- **Objective 13:** K-Means — Clustering games by Total Reviews, Average Playtime, Positive Review Rate, Helpfulness, and Humor—Which game segment retains players best?

4 Dataset and Data Processing Methodology

4.1 Data Source and Rationale

The data utilized in this report is derived from the Steam Games, Reviews, and Rankings dataset on Kaggle. This dataset comprises over 990,000 rows extracted directly from the Steam platform, focusing on the Top 40 leading games by sales, revenue, and reviews across six core genres: Action, Adventure, Role-Playing, Strategy, Simulation, and Sports & Racing.

The data collection process strictly adhered to Steam's *robots.txt* guidelines. A total of 290 games were compiled for information and ranking. However, the user review segment contains only 242 games, as certain titles containing sensitive content (nudity) were filtered out to ensure data appropriateness.

4.2 Data Tables Description

The raw dataset consists of three files, each containing specific types of information:

- **games_description.csv**: Provides detailed specifications of the games. Data fields include: game title (*name*), brief/detailed description (*short_description, long_description*), genre (*genres*), minimum/recommended system requirements (*minimum/ recommend_system_requirement*), release date, developer (*developer*), publisher (*publisher*), overall player rating (*overall_player_rating*), total reviews from purchasers, total English reviews, and the URL to the Steam page.
- **steam_game_reviews.csv**: Stores actual player feedback. Fields include: review content (*review*), total hours played (*hours_played*), helpful votes (*helpful*), funny votes (*funny*), recommendation decision (*recommendation*), review date (*date*), reviewed game name (*game_name*), and the user's name (*username*).
- **games_ranking.csv**: Records game rankings. Fields include: game name (*game_name*), genre (*genre*), ranking criteria (*rank_type*: sales, revenue, reviews), and the specific rank position (*rank*).

4.3 Data Model

The data model is designed as a normalized schema, utilizing the core entity **game_name** (Text type) as the primary key to link the tables.

- **games_description (1) – (N) steam_game_reviews**: A one-to-many relationship; a single game can possess thousands of reviews.
- **games_description (1) – (N) games_ranking**: A one-to-many relationship; a single game can appear across multiple ranking charts.
- **games_description_cleaned (1) – (1) cluster_input**: A one-to-one (1-1) relationship. This is an auxiliary table containing statistical features and cluster labels from the K-Means algorithm, facilitating the addition of segmentation attributes while preserving the integrity of the original data.

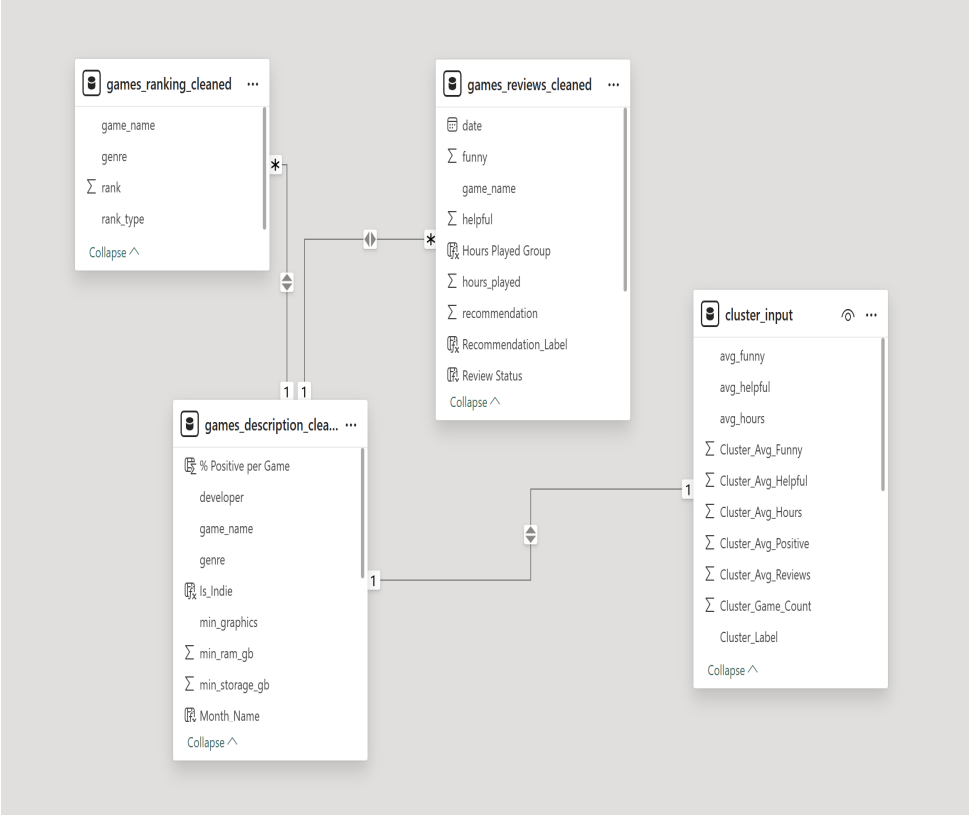


Figure 4.1: Entity-Relationship Diagram (ERD) of the Steam Dataset

4.4 Data Collection, Construction, and Preprocessing Methodology

4.4.1 Data Collection Process

The entirety of the initial raw data was downloaded directly from the Kaggle platform. At this stage, the data contained significant noise, formatting discrepancies, and vast free-text fields unsuitable for quantitative analysis in Power BI.

4.4.2 Database Construction Process

The database was constructed by optimizing the columns. Purely textual information without numerical analytical value was eliminated to optimize storage capacity and accelerate the Data Model's query performance.

4.4.3 Data Preprocessing Process

The cleaning and transformation processes were executed independently for each file via the Pandas library (Python):

File *games_description.csv*:

- *Data Omission:* Unnecessary text fields (*short_description, long_description, recommend_system_requirement, link*) were removed.
- *Hardware Specification Extraction:* The *minimum_system_requirement* column was parsed into three standard quantitative variables: *min_ram_gb*, *min_storage_gb*, and *min_graphics*.
- *Information Standardization:* The *genre* column was updated to synchronize with the ranking table. Release dates were standardized to the *YYYY-MM-DD* format. For *developer* and *publisher*, only the first name was retained if the list contained multiple values.
- *Data Type Casting:* Extraneous characters (such as commas, "% of..." strings) were stripped from review count columns and cast to Integer types.

File *games_ranking.csv*:

- Cleansed the *game_name* column by removing superfluous double quotes (") at both ends of the string, if present.
- Genre duplication handling: If a game possessed multiple differing *genre*s, the system automatically calculated and retained only the single best representative *genre* (based on the highest average ranking).

File *steam_game_reviews.csv*:

- *Data Omission:* The *review* (text content) and *username* columns were deleted to reduce model load.
- *Metric Standardization:* Thousands-separator commas were removed from the *helpful, funny*, and *hours_played* columns, and converted to Integer formats.
- *Sentiment Digitization:* The *recommendation* column was transformed into a binary variable: "Recommended" $\rightarrow$ 1, and "Not Recommended" $\rightarrow$ 0.
- *Time Standardization:* The *date* column was converted to the *YYYY-MM-DD* format.

5 Data analysis and visualization

Dashboard Steam Gaming Market - Object 1, 2, 3

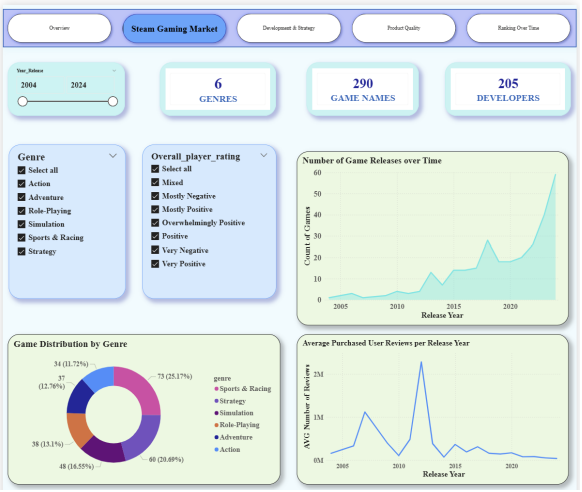


Figure 5.1: Dashboard Steam Gaming Market 1

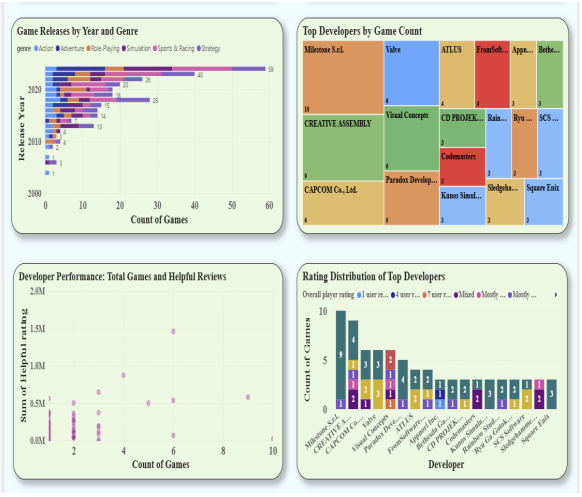


Figure 5.2: Steam Gaming Market 2

5.1 Object 1: How does the number of games released on Steam change each year, and has the market become saturated yet?

5.1.1 Charts and methodology

The comparative analysis method is used to clarify the relationship between supply and demand on Steam. The Area Chart is used to show the number of games released each year, reflecting the growth of supply. Meanwhile, the Line Chart presents the average number of user reviews, representing the level of interest and engagement from players.

5.1.2 Analysis results

The results show that the number of games released started to increase rapidly from 2015 and reached its peak in 2024, indicating a strong growth in supply. In contrast, the average level of player engagement peaked around 2012–2013, then decreased significantly and remained at a low level, with no clear recovery in recent years.

5.1.3 Comments & Conclusion

The contrast between these two trends shows that supply is increasing rapidly while demand does not grow at the same rate, leading to fragmented player attention. As a result, new games find it harder to achieve high levels of engagement compared to the past. It can be concluded that although the Steam market is still growing in terms of the number of products, it has entered a stage of interaction saturation, which increases competition among developers.

5.2 Object 2: Which game genres account for the largest share, and how has the distribution of genres changed over time?

5.2.1 Charts and methodology

To address this objective, the study combines structural analysis and trend analysis, while also making full use of dynamic cross-filtering features through slicers in Microsoft Power BI.

The Donut Chart is used to show the overall distribution of game genres, helping to quickly identify the dominant market segments. The Stacked Bar Chart is applied to track changes in the number of games for each genre over time. This combination allows a more detailed view of the lifecycle and popularity of each genre at different periods.

5.2.2 Analysis results

The results show that two genres clearly dominate the market: Sports & Racing with 25.17% (73 games) and Strategy with 20.69% (60 games), while genres such as Action and Adventure have a lower share. This trend is reinforced by the time-based chart, where most of the increase in the number of games in recent years is concentrated in these two genres.

5.2.3 Comments & Conclusion

The data reflects that players tend to prefer genres that require strategic thinking or offer strong competition, rather than traditional action experiences. This trend is not only stable but also continues to grow over time. Therefore, Sports & Racing and Strategy play a key role as dominant segments, while Action and Adventure show signs of saturation.

Developers should consider focusing on these stronger genres to increase their chances of success in the market.

5.3 Object 3: Which developers contribute the largest number of games, and what is the quality of their games?

5.3.1 Charts and methodology

To address this objective, the study applies market fragmentation analysis combined with multi-dimensional correlation analysis. Three types of charts are used: a Treemap to show the scale and fragmentation of developers, a Stacked Column Chart to evaluate product quality based on user ratings, and a Scatter Plot to measure community engagement through the total number of helpful votes

5.3.2 Analysis results

The Treemap results show that the market is highly fragmented, with 205 developers producing 290 games, most of which are small in scale. Some developers, such as Milestone S.r.l and CREATIVE ASSEMBLY, lead in terms of the number of games, but they do not dominate the market completely. The stacked column chart indicates that most of their products receive positive reviews. However, the scatter plot shows that developers with a large number of games do not always achieve the highest level of engagement. In contrast, developers with a moderate number of games can gain a much higher number of helpful votes.

5.3.3 Comments & Conclusion

Developers on Steam follow very different strategies. Focusing on producing many games may increase visibility, but it is not enough to build a strong and loyal community. In conclusion, the key to success on this platform is that quality is more important than quantity. Developers who focus on a moderate number of well-developed games are more likely to attract loyal players, create long-term value, and build a stronger community compared to those who release many games without enough depth.

Dashboard Development & Strategy - Objective 4, 5, 6

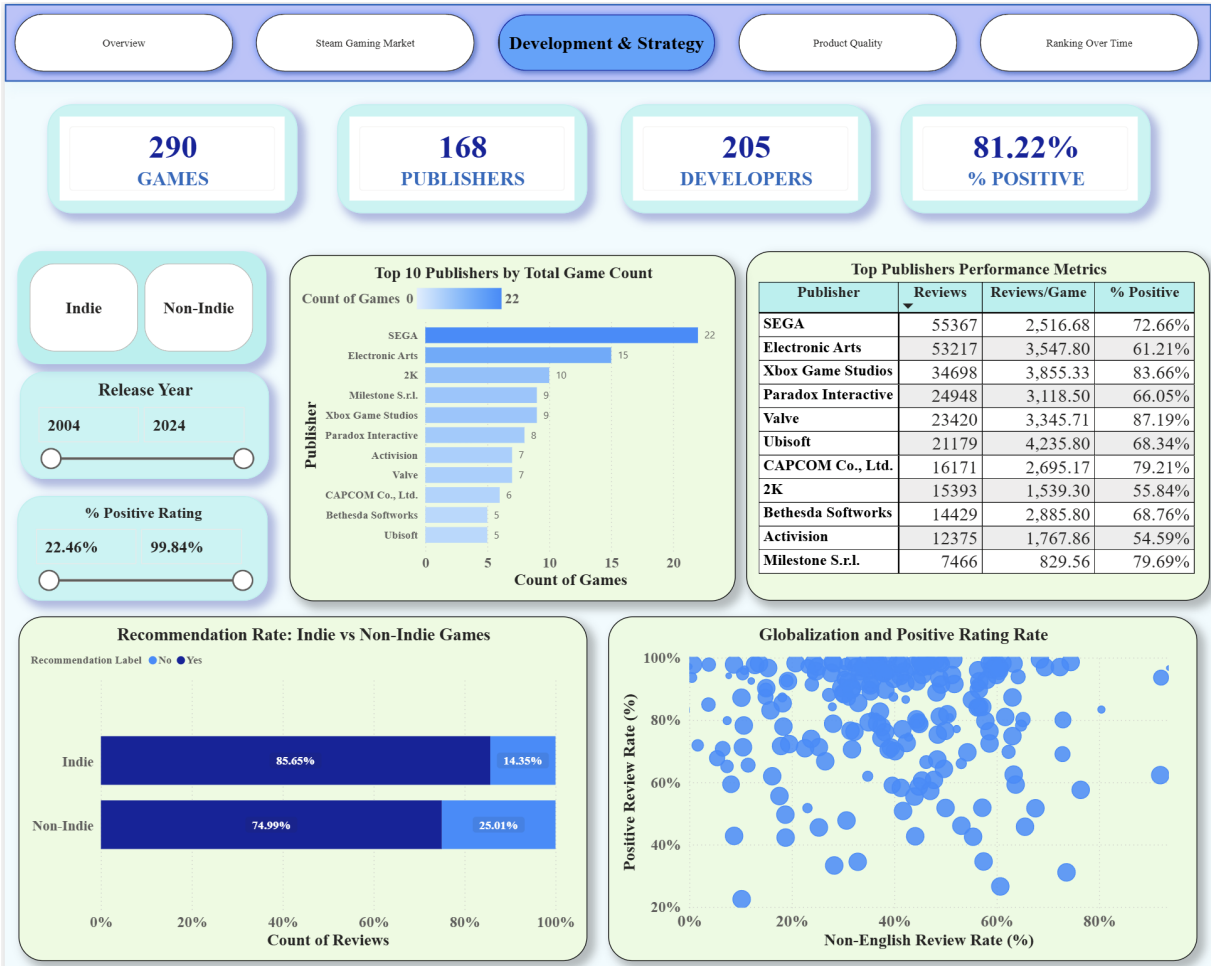


Figure 5.3: Dashboard Development & Strategy

5.4 Objective 4: Which publishers have the most games and the highest total number of reviews, and who is dominating Steam?

5.4.1 Charts and Methodology

To address this objective, the report utilizes a combination of a **Clustered Bar Chart** and a **Matrix Table**. The horizontal clustered bar chart facilitates the visualization and rapid identification of the Top 10 leading publishers in terms of product volume. Accompanying this, the matrix table serves as a detailed data cross-reference, providing additional dimensional metrics regarding the total number of reviews, the average number

of reviews per game, and the recommended rating percentage (% Positive). This enables an assessment of the actual quality behind the volume of releases.

5.4.2 Analysis Results

The results indicate that **SEGA** and **Electronic Arts (EA)** are the two leading publishers in terms of game volume (with 22 and 15 games, respectively). However, quantity does not strictly correlate with a high positive rating percentage. **Valve**, despite only making the top list with 7 titles, possesses the highest positive rating percentage among the group at 87.19% and an exceptionally high average number of reviews per game (3,346 reviews/game). In contrast, despite their large number of games, **2K** and **Electronic Arts** only maintain a moderate satisfaction rate, hovering around 55.84% and 61.21%, respectively.

5.4.3 Comments & Conclusion

The data reflects a stark contrast in the strategies of major entities on Steam. Companies like SEGA and EA lean towards a market-saturation strategy based on product volume. Meanwhile, publishers like Valve focus on a "quality over quantity" approach, ensuring that every released product achieves a high degree of polish and garners substantial support from the community.

5.5 Objective 5: How do indie games and publisher-backed games differ in terms of user engagement and recommended review rates on Steam?

5.5.1 Charts and Methodology

A proportional analysis method is applied utilizing a **100% Stacked Bar Chart**. This chart is the optimal choice for a direct comparison of the percentage distribution of **Recommendation (Yes)** and **Not Recommendation (No)** reviews between the two release categories (**Indie** and **Non-Indie**). This approach helps eliminate statistical skew caused by the variance in the absolute total number of reviews between the two groups, thereby providing a fair perspective on quality.

5.5.2 Analysis Results

The analysis demonstrates that Indie games (self-developed and self-published) record an overwhelming positive rating rate, reaching 85.65%. Conversely, the Non-Indie game group (released via third-party publishers) exhibits a significantly lower positive rating rate, settling at 74.99%.

5.5.3 Comments & Conclusion

The Steam player community exhibits a higher degree of preference and satisfaction towards independent gaming projects. The advantage of the Indie group may stem from the fact that these studios frequently maintain direct interaction and deep engagement with their community. Furthermore, they are more flexible in experimenting with innovative gameplay mechanics, and player expectations are generally set closer to the realistic scale of the project compared to blockbuster titles from major publishers.

5.6 Objective 6: Is the level of game globalization (measured by the proportion of non-English reviews) positively correlated with the total number of reviews and the positive rating rate on Steam?

5.6.1 Charts and Methodology

To identify correlations between quantitative variables, the report employs a **Scatter Plot**. The horizontal axis (X-axis) represents the degree of globalization (% Non-English), while the vertical axis (Y-axis) represents the satisfaction level (% Positive). The distribution of data points across this coordinate system aids in detecting trend clusters and evaluating whether the proportion of international players impacts the overall quality of the experience.

5.6.2 Analysis Results

The scatter plot illustrates that data points are densely clustered in the upper half of the vertical axis (% Positive > 60%) and span continuously from 0 to 80% on the horizontal axis (% Non-English). There is no distinct linear trendline demonstrating that an increase or decrease in the proportion of multilingual players alters the positive rating rate. Games achieving exceptional quality (approx. 90% - 100% Positive) appear evenly across both

the predominantly English-speaking game group and the group with a large international community.

5.6.3 Comments & Conclusion

The results confirm that language is neither a barrier nor a determining factor in a game's review quality. A well-designed product will consistently receive community recognition regardless of language. Therefore, developers should confidently implement localization strategies to scale their reach within the global market, without concern that cultural or linguistic differences will negatively impact the project's overall rating.

Dashboard Product Quality - Objective 7, 8, 9, 10

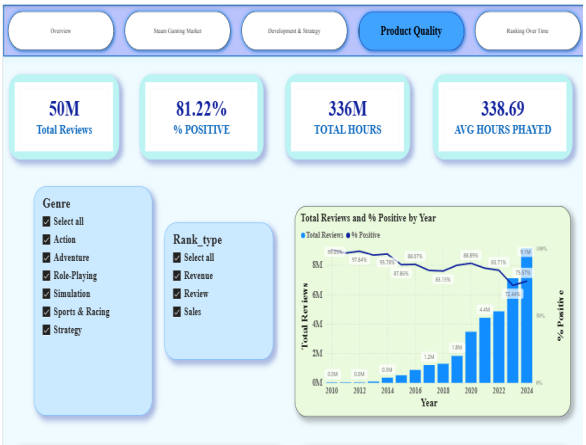


Figure 5.4: Market Overview and Yearly Review Trends

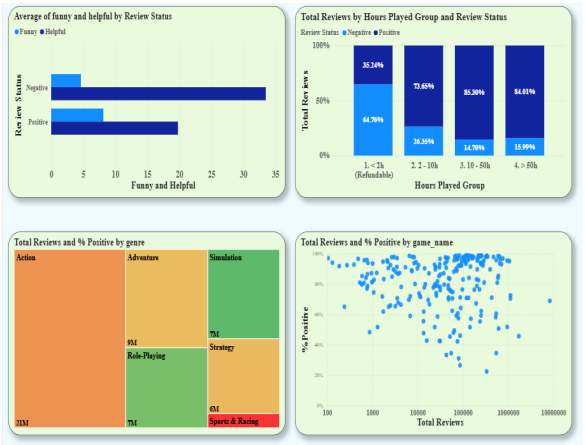


Figure 5.5: Review Analysis by Sentiment, Genre, and Volume

5.7 Objective 7: Trends in Total Reviews and Positive Rates Over the Years

5.7.1 Charts and Methodology

The objective is to analyze the fluctuation in the number of reviews and the positive review rate over the years to determine whether the gaming community is becoming stricter and more demanding as the market expands.

To achieve this, a **comparative analysis method** is applied using a Combo Chart (Bar and Line). The Bar Chart displays the total number of reviews (Total Reviews), reflecting the scale of community participation. Meanwhile, the Line Chart represents the positive review rate (% Positive), indicating player satisfaction levels and standards.

5.7.2 Analysis Results

The data shows a massive boom in total reviews, accelerating from 2014 and continuously surging between 2019–2024. In contrast, the positive review rate tends to decline. From nearly 100% in the 2010–2012 period, this rate steadily dropped to around 70–75% by 2023–2024.

5.7.3 Comments & Conclusion

The more the review volume booms, the more the positive rate decreases. As the player base expands and the gaming market grows, product evaluation standards become stricter. Gamers are genuinely becoming more demanding; a growing community brings higher standards, forcing developers to manage their output quality more rigorously to satisfy mass evaluation.

5.8 Objective 8: Analyzing Different Trends Between Negative and Positive Reviews

5.8.1 Charts and Methodology

The objective is to analyze the community's feedback behavior and engagement levels regarding different streams of opinion, based on playtime and average votes (Helpful/Funny). This clarifies player review trends and the impact of praise or criticism on the community.

Two charts are used: A Stacked Column Chart showing the proportion of Positive/Negative reviews across playtime groups (under 2 hours, 2–10 hours, 10–50 hours, over 50 hours); and a Bar Chart comparing the average "Helpful" and "Funny" votes between the two review sentiments.

5.8.2 Analysis Results

Data from the Stacked Column Chart reveals that under the 2-hour mark (the refund window), Negative reviews strictly dominate (around 65%). As playtime increases (2–10 hours and 10–50 hours), the negative rate drops sharply, with positive reviews making up over 80%. However, for the over 50 hours group, the Negative rate shows a slight uptick.

Regarding engagement, the Bar Chart indicates that Negative reviews receive an overwhelming number of "Helpful" votes (averaging over 33 votes/post) compared to Positive ones (around 20 votes/post). Conversely, Positive reviews attract more "Funny" votes.

5.8.3 Comments & Conclusion

Players are highly impatient with poor initial experiences, often dropping the game within the first 2 hours to secure a refund. Those who stick around longer are generally satisfied, but they may still turn away if the "end-game" content fails to retain them.

The community tends to value poor reviews that analyze game flaws over praise. Early Negative reviews heavily influence the community, meaning developers must optimize and provide a good initial experience to encourage player retention.

5.9 Objective 9: Analyzing Player Satisfaction by Game Genre

5.9.1 Charts and Methodology

The objective is to analyze the scale of engagement and player satisfaction for each game genre based on total reviews and positive review rates, clarifying how players perceive different genres in the gaming market.

A **Treemap** is used as the primary visualization method. The size of each block represents the total number of reviews (Total Reviews), reflecting the scale and appeal of each genre. The color of the block is conditionally formatted based on the positive review rate (% Positive), transitioning from Red/Orange (low quality) to Green (high quality, well-liked).

5.9.2 Analysis Results

The chart data indicates a clear divergence between market share and experience quality. The "Action" genre has the largest scale, completely dominating other groups in

player count; however, this block is colored Orange, showing that community satisfaction is only at a lower-average threshold.

In contrast, genres like "Simulation" and "Role-Playing", despite having smaller market shares, stand out in player satisfaction, proving they maintain very high praise rates. Most notably, the "Sports & Racing" group — possessing the smallest review count — is colored bright Red, representing rock-bottom negativity.

5.9.3 Comments & Conclusion

The overall picture shows that market size is not directly proportional to product satisfaction. While the Action genre attracts massive engagement, it is difficult to satisfy the community due to gamers' high standards and expectations.

Meanwhile, Role-Playing Games (RPG) and Simulation games genuinely meet customer needs, thereby providing good experiences for the gaming community. The Sports & Racing genre, on the other hand, is in a major crisis, with low engagement and many negative reviews. Developing Action games based on trends brings massive interaction but requires meeting stricter needs and standards. Conversely, carefully investing in groups like Simulation or RPG is a safer route and better received by the community.

5.10 Objective 10: Correlation Between Engagement Volume and Satisfaction Level

5.10.1 Charts and Methodology

The objective is to analyze the relationship between the actual review volume and experience quality based on engagement data for specific titles. This clarifies whether games attracting massive attention can maintain high satisfaction levels.

To achieve this, a **Scatter Chart** is used. The horizontal axis displays the review volume from verified players on a logarithmic scale to easily observe the broad data spectrum. The vertical axis (Y-axis) represents the positive review rate (% Positive). Each point represents a specific title (game_name), with the size indicating the total number of reviews (Total Reviews).

5.10.2 Analysis Results

Games with a relatively low review count (under 10,000) achieve very high positive rates; over half of these games surpass the 80% positive mark.

For games with massive engagement (100,000 – 1,000,000 reviews), the points scatter widely along the vertical axis. The positive rate for this group fluctuates wildly, with many dropping to 40% – 60%, and some even hitting the 20% threshold. However, a large group of games still maintains a 90% – 100% positive rate.

5.10.3 Comments & Conclusion

The larger the engagement scale, the harder it is to ensure mass satisfaction. Games with fewer reviews often serve a niche community, making it easier to satisfy their core player base. However, when a game becomes popular, it faces significant differences in expectations, preferences, and stricter standards from diverse player groups.

The fact that a large group of titles still maintains the 90% – 100% positive mark proves that true excellence can still win over the masses. A large review volume acts as a "filter," exposing the flaws of mid-tier games while affirming the quality of true masterpieces.

Dashboard Ranking Over Time - Objectives 11, 12, 13

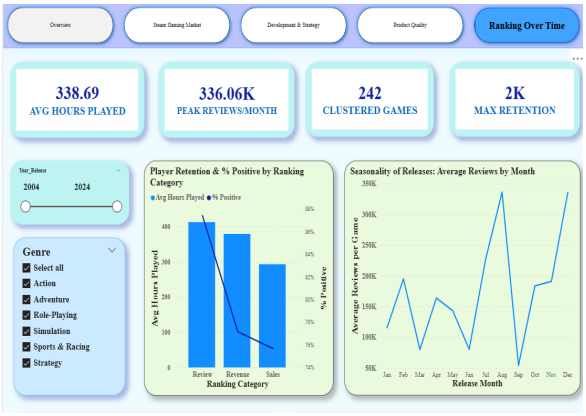


Figure 5.6: Ranking and Seasonality Analysis

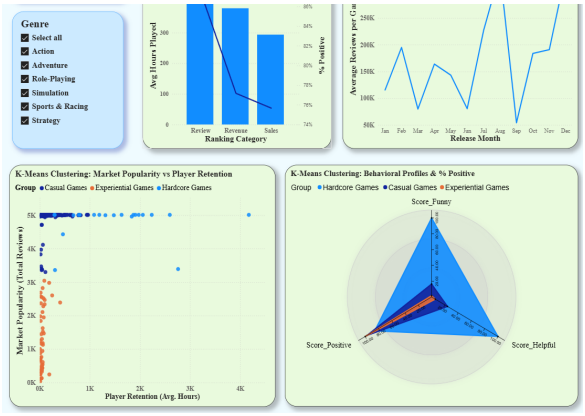


Figure 5.7: K-Means Clustering Distribution and Profiles

5.11 Objective 11: Do top-ranking games by Revenue have higher average playtime and recommendation rates compared to other ranking types (Sales/Review)?

5.11.1 Charts and methodology

The objective is to compare the player experience quality between commercial ranking categories (Revenue, Sales) and the community-driven category (Review) through two core metrics: average playtime and positive review rate. This analysis aims to clarify whether financial success aligns with long-term player retention and user satisfaction.

To achieve this, a **comparative analysis method** is applied using a **Line and Clustered Column Chart**. The columns represent the Average Hours Played for each Ranking Category, while the line represents the percentage of positive reviews (% Positive). This allows for a synchronized observation of the relationship between business performance and actual community feedback on a single coordinate axis.

5.11.2 Analysis Results

The results indicate a clear differentiation among the three ranking groups. The **Review** group leads absolutely in average playtime at approximately 400 hours, significantly outperforming the Revenue group (380 hours) and the Sales group (290 hours). Regarding the positive review rate, a similar trend is recorded: the **Review** group possesses the highest % Positive rate (approx. 87%), while the **Revenue** and **Sales** groups have significantly lower rates, ranging from 75% to 77%.

Notably, games in the **Top Revenue ranking do not achieve leading positions** in either quality metric. This demonstrates that high revenue is not a guaranteed indicator of player satisfaction or long-term engagement.

5.11.3 Comments & Conclusion

The analysis confirms that **commercial ranking (Revenue/Sales) and experience quality are two independent axes** in the Steam market. Being a top-seller or high-revenue generator does not equate to a game's ability to retain players long-term or receive high positive acclaim. Conversely, the group of games highly endorsed by the community (Top Review) exhibits the strongest indicators of experience quality and player loyalty.

This suggests that developers should not focus solely on short-term strategies to boost sales figures. Investing in core gameplay quality and user experience—values reflected through Review metrics—is the sustainable path to building a brand and a loyal player community on the Steam platform.

5.12 Objective 12: Which month of the year receives the most reviews—is there a 'golden season' for releases?

5.12.1 Charts and methodology

The objective is to analyze the time seasonality and cycles in game release activities on Steam to determine if there are specific times of the year when a new title can receive significantly higher community interaction and virality compared to other periods.

To achieve this, a **time cycle analysis method** is applied through a **Line Chart**. The horizontal axis represents the 12 months of the year (Release Month), while the vertical axis represents the **average reviews received per game** (Average Reviews per Game). Approaching via the average instead of the total helps eliminate deviations caused by the unequal number of games released across months, thereby accurately reflecting the level of reception and attraction of products at the time of launch.

5.12.2 Analysis Results

The chart shows a clear seasonal trend with three prominent features. First, **August and December** record the highest average review scores—reaching approximately **336.06K reviews/game** (the peak for the entire year). Second, the chart records a **strong growth trend during the summer months**: starting from a low point in June, interaction volume spikes in July and reaches its peak in August. Conversely, **March, June, and especially September** are identified as the "valleys" of the release cycle, when average interaction levels fall to their lowest points.

5.12.3 Comments & Conclusion

The data confirms the existence of two **"golden seasons" for game releases on Steam**. Besides the year-end peak associated with festivals, the **summer period (June to August)** shows steady growth. This reflects the impact of the summer break, when young players

have more time to experience games, creating a motivation that drives massive interaction for titles released during this season.

However, this is also the period of fiercest competition as major publishers focus on launching products. For smaller studios, choosing to release in months with decent interaction but less competition (such as April or October) could be a wise strategy to avoid being overshadowed by the massive media campaigns of titles from famous developers during peak seasons.

5.13 Objective 13: K-Means — Clustering games by Total Reviews, Average Playtime, Positive Review Rate, Helpfulness, and Humor—Which game segment retains players best?

5.13.1 Charts and methodology

The objective is to cluster titles based on behavioral characteristics and multi-dimensional player interactions to identify distinct Market Segments.

To achieve this, an **unsupervised machine learning method (K-Means Clustering)** was applied using five standardized input variables. *Special note:* The `total_reviews` variable in this model is defined as the **total number of reviews possessing full information** regarding attitude (*Positive*), helpfulness (*Helpful*), and humor (*Funny*) extracted from the cleaned dataset (`games_reviews_cleaned.csv`). This selection ensures that the K-Means algorithm is trained on data with the highest levels of actual interaction and knowledge-sharing quality.

Execution process: The entire algorithm was run via a Python script within Power Query, outputting results into a specialized table named `cluster_input`. This table stores cluster labels (`Cluster_Label`) along with summary statistics for each segment, established in a **one-to-one (1-1)** relationship with the original table to extend segmentation attributes.

The process utilized Python integration in Power Query with the parameter `n_clusters=3`. The resulting cluster labels were merged back into the original dataset to facilitate visualization through a **Scatter Plot** (Popularity vs. Retention distribution) and a **Radar Chart** (Behavioral Profiles).

5.13.2 Analysis Results

Three segments were formed with distinct behavioral characteristics:

- **Hardcore Games (Grinding Segment):** This group retains players best with superior average playtime, with certain titles recording engagement levels exceeding 4,000 hours. Despite a small sample size, this community is extremely vibrant, leading absolutely in interaction scores (`Score_Helpful` and `Score_Funny`). However, it is also the most "demanding" group, possessing the lowest positive review rate (`Score_Positive`) among the three clusters.
- **Casual Games (Mainstream Segment):** Accounts for a large proportion of the dataset with extremely high *Market Popularity*. Players in this group are generally satisfied with their experience (`Score_Positive` is relatively high), but engagement levels remain average, with less tendency for deep interaction or contributing long-form content to the community.
- **Experiential Games (Experience Segment):** This is the "quality over quantity" group. Despite having the lowest playtime and review volume, this segment possesses the **highest positive review rate**. This reflects the nature of story-driven or short-term experience games: players are highly satisfied but leave after completing the game.

To clarify the profiles of each cluster, the report summarizes the actual average statistics in Table 5.1:

Table 5.1: Actual average statistics of the segments

Game Segment	Hours	Reviews	Positive	Helpful	Funny
Hardcore Games (Cluster 1)	1509.89	4823	0.72	74.18	31.82
Casual Games (Cluster 0)	213.07	4930	0.82	17.76	5.21
Experiential Games (Cluster 2)	55.70	1252	0.84	4.11	0.55

Normalization: Due to variables having vastly different scales, the group applied a normalization technique based on maximum values. This formula brings all indices to a 100-point scale to compare relative strengths between clusters on the radar chart:

$$Score = \frac{\text{Cluster Value}}{\text{Column Maximum Value}} \times 100 \tag{5.1}$$

The normalized score table (Table 5.2) is used to plot the Radar chart representing interaction behavior:

Table 5.2: Normalized scores (0-100) for the Radar Chart

Game Segment	Score Positive	Score Helpful	Score Funny
Hardcore Games	85.71	**100.00**	**100.00**
Casual Games	97.62	23.94	16.37
Experiential Games	**100.00**	5.54	1.73

5.13.3 Comments & Conclusion

Analysis from the K-Means model reveals a **key paradox**: the group with the highest satisfaction rate (*Experiential*) has the lowest player retention, while the group with the lowest satisfaction rate (*Hardcore*) is the most deeply engaged and generates the greatest community interaction value.

This confirms that % Positive is not the sole measure of success on Steam. Developers must decide: invest in deep gameplay to build a sustainable **Hardcore** community, or choose the **Experiential/Casual** model to prioritize brand reputation and low risk.

6 Summary of Dashboard and Key Insights

After conducting a series of analyses across 13 objectives on the Steam dataset, the group has derived several valuable core insights into the current gaming market. These findings not only reflect the present situation but also suggest important strategic directions for game developers:

1. **Interaction Saturation in the Market:** Although the number of game releases has grown to record levels over the years (peaking in 2024), the average level of player engagement has shown a declining trend. This indicates a fragmentation of attention; launching a new game today finds it increasingly difficult to capture concentrated interest compared to earlier periods.

2. **The Advantage of Indie Games:** *Indie studios* demonstrate strong community engagement with a dominant Positive rate (85.65% compared to 74.99% for Non-Indie).

This confirms that a strategy focused on quality experience and direct interaction with players is more effective than mass production.

3. **Revenue Does Not Equal Retention:** Games with the highest revenue (*Top Revenue*) are not necessarily those that retain players the longest. Games that receive the highest community ratings (*Top Review*) are the true representatives of user loyalty and satisfaction.

4. **Golden Release Seasons:** December and August are two strategic periods with explosive engagement levels. In addition, the summer period from June to August is also a highly potential time for game developers.

5. **Paradox from the K-Means Model:** The *Hardcore* group (retaining players for more than 1,500 hours) contributes the highest level of community engagement but is also the most demanding in terms of evaluation. In contrast, the *Experiential* group delivers the strongest brand reputation with an extremely high satisfaction rate despite shorter engagement lifecycles.

Conclusion: To succeed on Steam, developers need to choose between building a sustainable ecosystem through deep gameplay (*Hardcore*) or optimizing user emotions over a shorter period (*Experiential*). In a context of excessive supply, understanding player behavior and delivering high product quality are the decisive factors for a game's survival.

7 Video Demo

Click here to watch the video

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